



AMIR HAMZAH ABD AZIZ

Data Engineer | Software Developer

Kuala Lumpur, Malaysia | amirium381@gmail.com | [+6018-9103791](tel:+6018-9103791) | Malaysian

LinkedIn: <https://www.linkedin.com/in/muhammad-amir-hamzah/>

Portfolio: <https://amoiir.github.io>

Professional Summary

A data-driven IT professional and AI Apprentice with a **First-Class Honours degree** and recipient of **Dean's List Award** for four consecutive years. Recently completed the **Gamuda AI Academy Full Stack AI Development** programme, gaining **2nd runner up award** and deep technical fluency in the **Google Cloud ecosystem** as well as **Generative AI implementation**.

Proven expertise in **optimizing ETL workflows**—demonstrated by processing **1.9M rows in three days** at Perodua—and building **AI-integrated applications** using Gemini API and RAG concepts. Proficient in **SQL, Python, and Power BI**, with a strong foundation in **business intelligence** and **automated data integration**. Available to start working **Immediately**.

Key Skills

CORE SKILLS:

Data Engineering & ETL | AI & LLM Implementation | Data Modelling & Visualisation | Agile Project Management | Software Development

SOFT SKILLS:

Strategic Technical Guidance | Stakeholder Communication | Bilingual Proficiency (Malay & English)

Specialised Technical Upskilling

Full Stack Development & AI Engineering (Immersive) | Gamuda AI Academy (in collaboration with Google Cloud), *Selangor* | January to March 2026

- Selected for Gamuda's pioneering initiative to foster an "AI-first" organizational culture, mastering the integration of AI into industrial and corporate workflows.
- **Cloud Architecture:** Leveraged Google Cloud Platform (GCP) and Vertex AI to build scalable, production-ready solutions using RAG (Retrieval-Augmented Generation).
- **Full-Stack Engineering:** Developed end-to-end applications using React.js (Frontend) and FastAPI/Python (Backend) to automate complex business processes.
- **Capstone Project:** Engineered a real-world AI solution demonstrating the ability to translate high-level strategic objectives into functional technical architecture.

Professional Experience

System Engineer (Big Data & Visualisation), Perodua Sdn Bhd, *Selangor* | April to October 2025
Managed enterprise-level data integration and automation within the digitalization department to support cross-company analytics.

- **Automated Data Pipelines:** Owned and maintained SSIS pipelines to automate Oracle-to-SQL Server ETL, achieving the transfer of 1.9M rows within 72 hours.
- **Centralized Data Integrity:** Implemented Master Data Services (MDS) to establish a "single source of truth" across multiple Perodua group companies.
- **Quality Assurance:** Enhanced data cleansing and matching using Data Quality Services (DQS) and led UAT bug-fixing to ensure production-ready pipelines.

Android Programmer (Internship), WADAH Technologies, *Melaka* | March to August 2024
Developed and optimized mobile features for a vehicle rental platform in an agile, data-driven environment.

- **Feature Engineering:** Built core Android features using Kotlin and MVVM architecture, improving overall app functionality and user experience.
- **API Integration:** Integrated REST and Google Maps APIs to enable location-based services and seamless transaction processing.

Education

First Class Honours

Bachelor of Information Technology, Specializing in Business Intelligence and Analytics

International Islamic University Malaysia, *Kuala Lumpur*, 2024

- Graduated 3.75 CGPA.
- Served as **KPMG Student Ambassador** for 2023.

Professional Certifications

- **Comptia Data+ ce**, CompTIA, 2025
- **SQL Fundamentals**, DataCamp, 2025
- **Intermediate Python**, DataCamp, 2025

Project Highlights

Airis: Camera-First Language Learning Application, Lead Developer | 2026

Led a 7-person cross-functional team to develop a full-stack mobile application focused on real-time image processing and AI-driven education. Conducted Focus Group Discussions with industry leaders like Pandai, Awantec and Gamuda Enabling Academy to ensure market relevance.

- **Award-Winning Development:** Secured a Top 3 ranking among 8 groups, receiving specific recognition for excellence in core image processing features.
- **Advanced AI Implementation:** Integrated Gemini API for bilingual object descriptions and applied Retrieval-Augmented Generation (RAG) to improve chatbot accuracy.

- **Technical Leadership:** Provided strategic guidance to non-technical stakeholders while managing the end-to-end development of the mobile platform.
- **Mentorship focus:** Mentored team members through rigorous code reviews, ensuring clean code and DRY principles to ensure enterprise-grade scalability and maintainability.

SwitchAR: IoT-Based Smart Home Automation, Final Year Project | 2024

Partnered project aimed at enhancing independence for the elderly through an Augmented Reality (AR) virtual switch control system. Developed a modern solution to help users interact with home appliances with minimal physical effort.

- **Cross-Platform Integration:** Proposed and built a mobile application utilizing Flutter and Unity to bridge the gap between AR concepts and IoT hardware.
- **User-Centric Design:** Focused on accessibility and ease of use, ensuring the interface met the specific needs of elderly users.

CRM Interactive Excel Dashboard, Data Analyst Project | 2025

Developed a high-performance dynamic dashboard to analyze sales efficiency and product performance. Focused on delivering actionable customer insights to enhance organizational decision-making processes.

- **Data Visualization:** Leveraged Pivot Tables and Pivot Charts to visualize complex sales datasets, improving the speed of performance reviews.
- **Efficiency Analysis:** Identified key trends in sales team efficiency, providing a clear overview of product-market fit and revenue drivers.

Technical Proficiencies

PROGRAMMING:

Python | SQL | Kotlin | JavaScript | Dart | Java | C++ | PHP | HTML | CSS

BI & DATA TOOLS:

Power BI | Tableau | Alteryx | Excel (Pivot Tables/Dashboards)

AI & DATABASES:

Generative AI (LLMs, RAG) | Microsoft SQL Server | PostgreSQL | Firebase | MongoDB

Awards

- **2nd Runner Up Award, Gamuda Cohort 5 Capstone Project, Gamuda AI Academy | 2026**
- **8 times Dean's List Award, International Islamic University Malaysia | 2020-2024**
- **Malaysia Data Innovation Talent 2023 (Top 50 among 120 groups), Department of Statistics Malaysia (DOSM), Bank Negara Malaysia, SAS Institute | 2023**

References are available upon request.